

Mahmoud Elseady

Software Engineer — Backend & Systems

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FOCUS .NET / C# · Network Design · Web Interfaces

01About

I'm a software engineer working mainly on the backend with C# and .NET, and I like understanding a system end to end rather than just the code layer — that's taken me into network design and, when a project needs it, the front end too. I'm finishing a computer engineering degree at the moment and use projects like the ones below to practice building things properly, from the data model up.

Comfortable moving between application code, network architecture, and interface work on the same project.

02Skills & Services

BACKEND	C#, .NET, object-oriented design, console & application logic, data modeling
NETWORKING	Network design in Cisco Packet Tracer — VLAN segmentation, inter-VLAN routing, IP addressing plans
FRONT-END	HTML, CSS, JavaScript — responsive layouts, interactive UI behaviour
SECURITY	Working knowledge of the OWASP Top 10 and web application testing fundamentals

03Selected Work

01 Digital Banking System

C# · CONSOLE APPLICATION

Academic project — sole developer

```
welcome to the Bank system
please enter the number of operation you want to perform:
1. Open Default Account
2. Open Current Account
3. Open Saving Account
4. find account by ID
5. Display all accounts
6. Display total bank balance
7. Exit
> 5
Account ID: 3  Account Type: Current  Balance: 4250  Owner: Sara Adel
```

FIG. 01 – actual console output from Program.cs

- Designed the account hierarchy with inheritance: a base *BankAccount* class extended by *CurrentAccount* (overdraft limit) and *SavingAccounts* (interest on deposit).
- Implemented customer records, deposits, withdrawals, transfers between accounts, and a per-account transaction log.
- Tracked total bank balance across every account using a shared static counter.

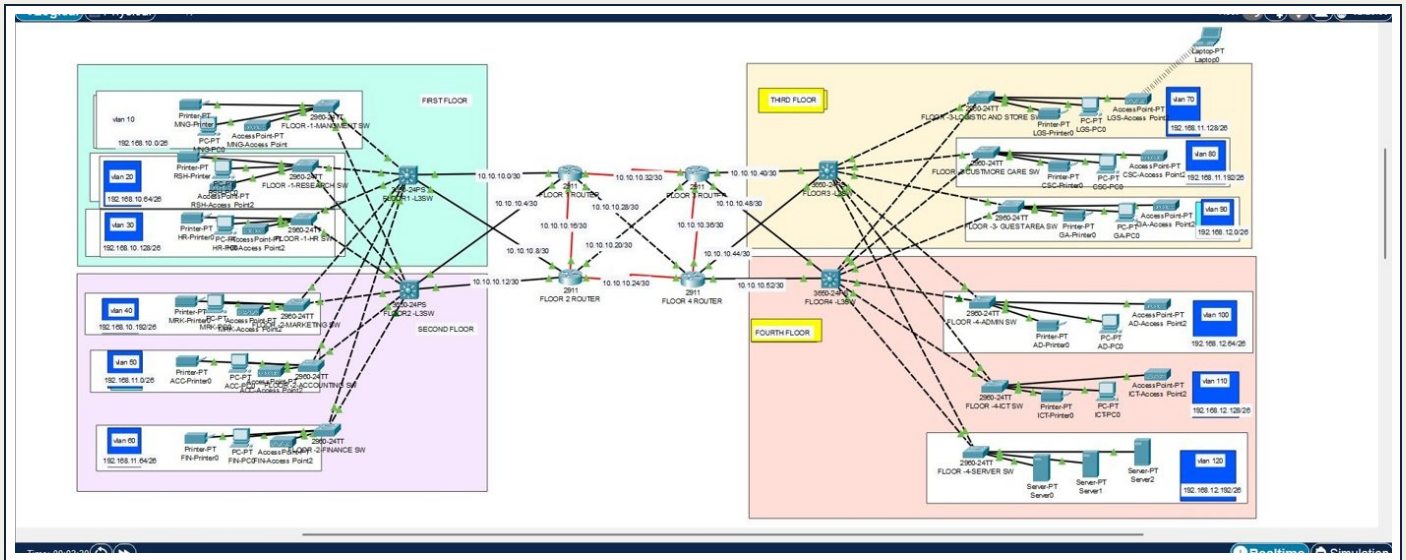


FIG. 02 – 4-floor branch topology, 11 department VLANs

- Laid out a 4-floor branch network with a dedicated VLAN per department — Management, Research, HR, Marketing, Accounting, Finance, Logistics, Customer Care, Guest Area, Admin, and ICT.
- Connected floors through L3 switches and routers, with a structured /30 addressing scheme on every router-to-router link.
- Planned a separate /26 subnet per VLAN, e.g. 192.168.10.0/26 for Management up to 192.168.12.192/26 for the server VLAN.

03 Login & Registration Interface

HTML · CSS · JAVASCRIPT

Front-end practice project

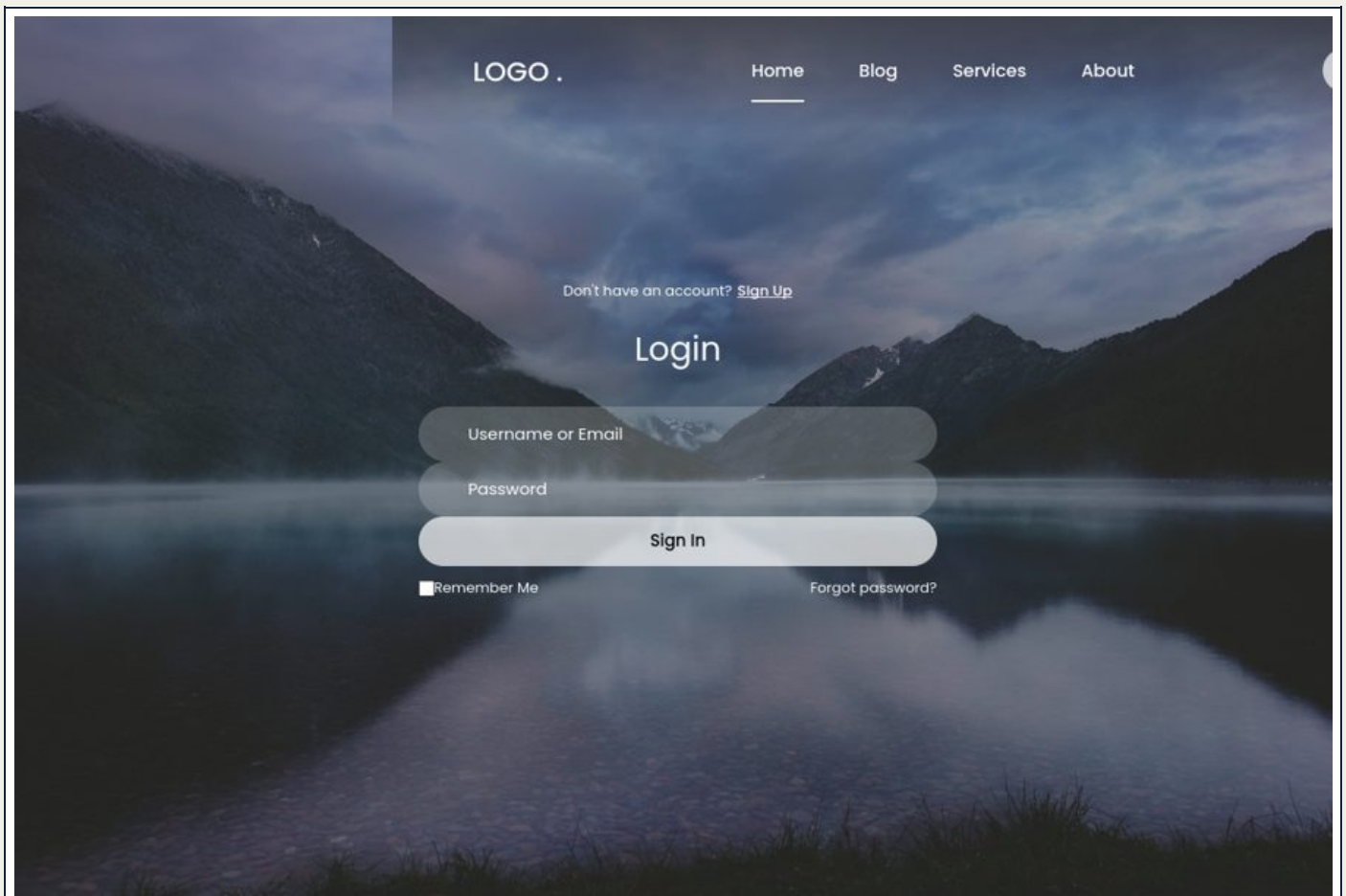


FIG. 03 – sliding login / sign-up panel

- Built a single page that slides between a login form and a sign-up form, driven by plain JavaScript — no framework.
- Styled with a full-bleed background photo, translucent input fields, and a collapsible menu for smaller screens.